Pontryagin-Based Solver with Universally Smoothed Hamiltonian, Adaptive ΔT, and PA-Bundle Refinement

March 15, 2026

Presentation Overview

Previous Meeting

To do:

- Fix the Symplectic Euler scheme:

$$\begin{aligned} \bar{X}_{n+1} &= \bar{X}_n + \Delta t H_\lambda(\bar{\lambda}_{n+1}, \bar{X}_n), \quad \bar{X}_0 = X_0, \\ \bar{\lambda}_n &= \bar{\lambda}_{n+1} + \Delta t H_x(\bar{\lambda}_{n+1}, \bar{X}_n), \quad \bar{\lambda}_N = g_x(\bar{X}_N) \end{aligned} \tag{1}$$

- Optimize the implementation for sparse matrix allocation and sparse solver for linear systems.
- Produce the plot t vs Δt and error density.

The Followign example is taken from Karlsson et. al. [1]:

Hypersensitive optimal control

Minimize

$$\int_0^{25} \left(X(t)^2 + \alpha(t)^2\right)\, \mathrm{d}t + \gamma\left(X(25) - 1\right)^2 \tag{2}$$

subject to

$$X'(t) = -X(t)^3 + \alpha(t), \qquad 0 < t \leq 25, \tag{3}$$

$$X(0) = 1. \tag{4}$$

for some large $\gamma > 0$.

Original time refinement criterion

The code builds a *local* indicator $\eta_{\text{time},i}$ per interval by measuring the variation of the smoothed Hamiltonian gradients. Then per interval $[t_i, t_{i+1}]$:

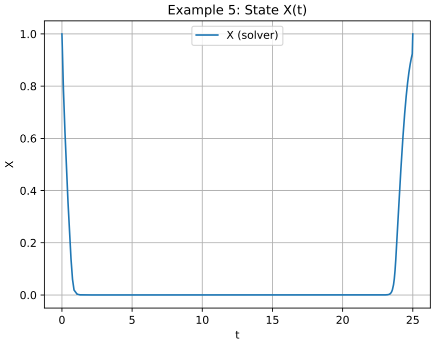
$$\eta_{\text{time},i} = \Delta t_i \Big(\|\nabla_p H_\delta(P_{i+1}, X_{i+1}, t_{i+1}) - \nabla_p H_\delta(P_i, X_i, t_i)\|_2 \tag{5}$$

$$+ \|\nabla_x H_\delta(P_{i+1}, X_{i+1}, t_{i+1}) - \nabla_x H_\delta(P_i, X_i, t_i)\|_2 \Big). \tag{6}$$

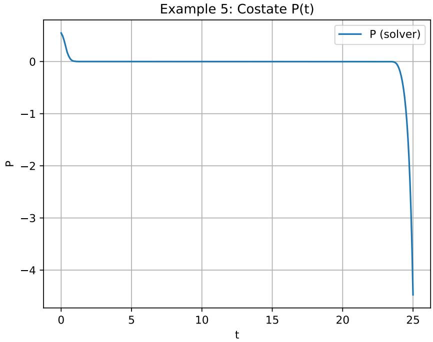
Global time indicator is the max:

$$\eta_{\text{time}} = \max_{i=0,\dots,N-1} \eta_{\text{time},i}. \tag{7}$$

Numerical Results



(a) Solution $\bar{X}$ of the discrete optimization problem (2) for $\varepsilon_{\text{time}} = 1e-1$, $\varepsilon_{\text{PA}} = 1e-3$, $\varepsilon_{\delta} = 1e-3$, $\delta_0 = 0.1$, $N_{\text{init}} = 30$.



(b) Solution $\bar{\lambda}$ of the discrete optimization problem (2) for $\varepsilon_{\text{time}} = 1e-1$, $\varepsilon_{\text{PA}} = 1e-3$, $\varepsilon_{\delta} = 1e-3$, $\delta_0 = 0.1$, $N_{\text{init}} = 30$.

Numerical Results

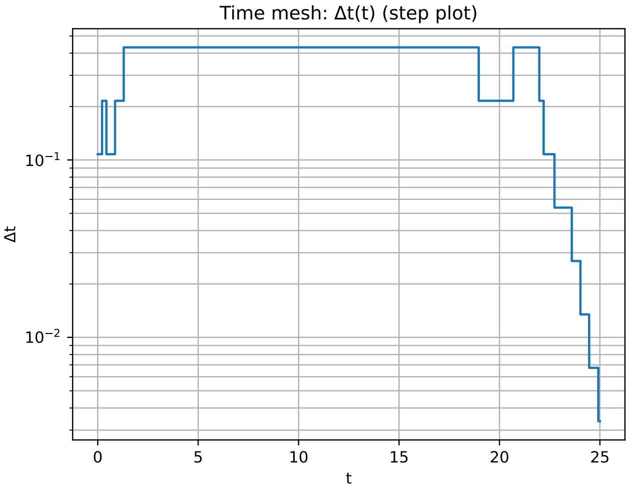


Figure: Mesh Δt for the optimization problem (2) for $\varepsilon_{\text{time}} = 1e - 1$, $\varepsilon_{\text{PA}} = 1e - 3$, $\varepsilon_{\delta} = 1e - 3$, $\delta_0 = 0.1$

Theorem (Karlsson et. al. [1])

Assume that all conditions in Theorem 2.3 are satisfied, that the Hamiltonian H is bounded in $C^2(\mathbb{R}^d \times \mathbb{R}^d)$, and that there exists a constant C such that

$$|\lambda_n - u_x(X_n, t_n)| \leq C\Delta t_{\max}, \quad \Delta t_{\max} := \max_n \Delta t_n \tag{8}$$

1. *The value function u is bounded in $C^3((0,T) \times \mathbb{R}^d)$.*
2. *There exists a neighborhood in $C([0,T], \mathbb{R}^d)$ around the minimizer $X : [0,T] \to \mathbb{R}^d$ of $u(x_0, 0)$ in which the value function u is bounded in C^3. Moreover*

$$\max_n |X_n - X(t_n)| \to 0 \quad \textit{as } \Delta t_{\max} \to 0.$$

If Condition 1 holds, then for every discretization $\{t_n\}$, the error $\bar{u}(x_0,0) - u(x_0,0)$ is

$$\bar{u}(x_0,0) - u(x_0,0) = \sum_{n=0}^{N-1} \Delta t_n^2 \rho_n + R, \quad \rho_n := -\frac{H_\lambda(X_n, \lambda_{n+1}) \cdot H_x(X_n, \lambda_{n+1})}{2} \tag{9}$$

$$|R| \leq C' \Delta t_{\max}^2,$$

If Condition 2 holds, then there exists a threshold time step Δt_{thres} such that for every discretization with $\Delta t_{\max} \leq \Delta t_{\text{thres}}$, the error representation (9) holds.

Theorem (Adaptivity)

Choose the error tolerance TOL, the initial grid $\{t_n\}_{n=0}^N$, and the parameters s and M, and repeat the following steps:

1. *Calculate $\{(X_n, \lambda_n)\}_{n=0}^N$ with the symplectic Euler scheme (1.7).*
2. *Calculate error densities $\{\rho_n\}_{n=0}^{N-1}$ and the corresponding approximate error densities*
$$\bar{\rho}_n := \operatorname{sgn}(\rho_n) \max(|\rho_n|, K\sqrt{\Delta t_{\max}}).$$
3. *Break if*
$$\max_n \bar{r}_n < \frac{\mathrm{TOL}}{N},$$
where the error indicators are defined by $\bar{r}_n := |\bar{\rho}_n| \Delta t_n^2$.
4. *Traverse through the mesh and subdivide an interval (t_n, t_{n+1}) into M parts if*
$$\bar{r}_n > s\frac{\mathrm{TOL}}{N}.$$
5. *Update N and $\{t_n\}_{n=0}^N$ to reflect the new mesh.*

We introduce a constant, $c = c(t)$, such that

$$c \leq \left| \frac{\bar{\rho}(t)[\text{parent}(n,k)]}{\bar{\rho}(t)[k]} \right| \leq c^{-1}, \quad t \in [t_n^{[k]}, t_{n+1}^{[k]}) \tag{10}$$
$$c \leq \left| \frac{\bar{\rho}(t)[k-1]}{\bar{\rho}(t)[k]} \right| \leq c^{-1}, \quad t \in [t_n^{[k]}, t_{n+1}^{[k]})$$

Theorem (Stopping)

Assume that c satisfies (10) for the time steps corresponding to the maximal error indicator on each refinement level and that

$$M^2 > c^{-1}, \qquad s \leq \frac{c}{M}. \tag{11}$$

Then, each refinement level either decreases the maximal error indicator with the factor

$$\max_n \bar{r}_n[k+1] \leq \frac{c^{-1}}{M^2} \max_n \bar{r}_n[k]$$

or stops the algorithm.

Theorem (Accuracy)

The adaptive Algorithm 2 satisfies

$$\limsup_{\mathrm{TOL}\to 0+}\left(\mathrm{TOL}^{-1}\left|u(x_0,0)-\bar{u}(x_0,0)\right|\right)\leq 1. \tag{12}$$

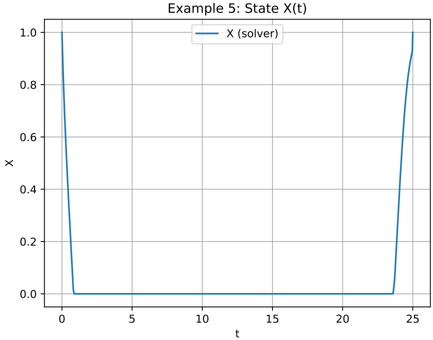
Theorem (Efficiency)

Assume that $c=c(t)$ *satisfies* (10) *for all time steps at the final refinement level and that all initial time steps have been divided when the algorithm stops. Then, there exists a constant* $C>0$*, bounded by* $M^2 s^{-1}$*, such that the final number of adaptive steps* N *of Algorithm 2 satisfies*

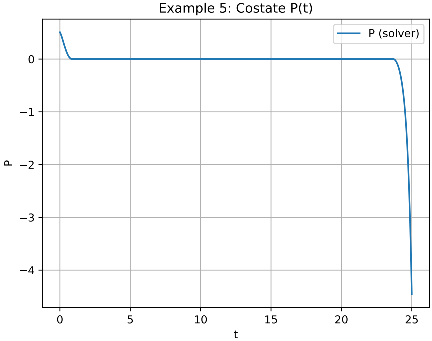
$$\mathrm{TOL}\,N\leq C\left\|\frac{\bar{\rho}}{c}\right\|_{L^{\frac{1}{2}}}\leq\|\bar{\rho}\|_{L^{\frac{1}{2}}}\max_{0\leq t\leq T}c(t)^{-1}, \tag{13}$$

and $\|\bar{\rho}\|_{L^{\frac{1}{2}}}\to\|\tilde{\rho}\|_{L^{\frac{1}{2}}}$ *asymptotically as* $\mathrm{TOL}\to 0^{+}$.

We consider the same example with $s = 0.25$ and $M = 2$ (since $c \approx 1$).



(a) Solution $\bar{X}$ of the discrete optimization problem (2) for $\varepsilon_{\text{time}} = 1e-1$, $\varepsilon_{\text{PA}} = 1e-3$, $\varepsilon_{\delta} = 1e-3$, $\delta_0 = 0.1$



(b) Solution $\bar{\lambda}$ of the discrete optimization problem (2) for $\varepsilon_{\text{time}} = 1e-1$, $\varepsilon_{\text{PA}} = 1e-3$, $\varepsilon_{\delta} = 1e-3$, $\delta_0 = 0.1$

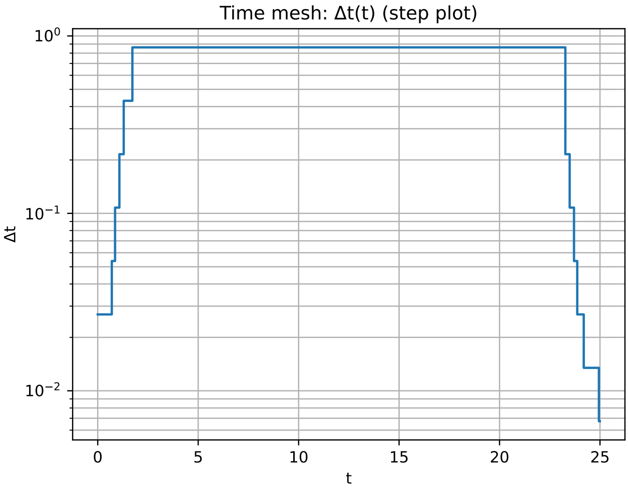
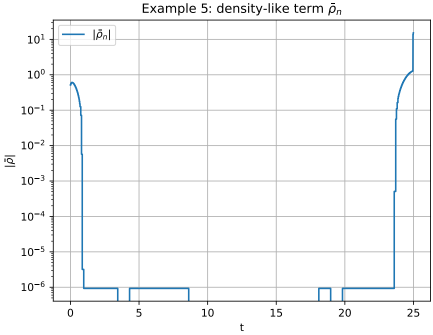
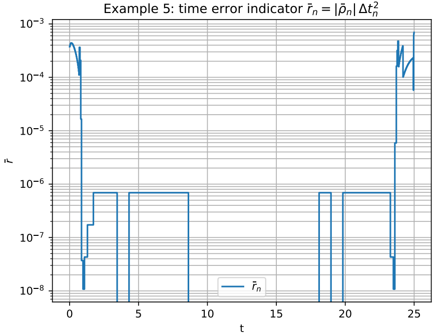


Figure: Mesh Δt for the optimization problem (2) for $\varepsilon_{\text{time}} = 1e-1$, $\varepsilon_{\text{PA}} = 1e-3$, $\varepsilon_{\delta} = 1e-3$, $\delta_0 = 0.1$

Numerical Results



(a) Error densities, $|\bar{\rho}|$ for $\varepsilon_{\text{time}} = 1e-1$, $\varepsilon_{\text{PA}} = 1e-3$, $\varepsilon_{\delta} = 1e-3$, $\delta_0 = 0.1$



(b) Error indicators $\bar{r}_n$‘ of the discrete optimization problem (2) for $\varepsilon_{\text{time}} = 1e-1$, $\varepsilon_{\text{PA}} = 1e-3$, $\varepsilon_{\delta} = 1e-3$, $\delta_0 = 0.1$

Metric	Value
Outer iterations (len(log))	16
Last outer iteration index	15
Number of time nodes ($\|\mathcal{T}\|$)	141
Number of intervals (N)	140
Bundle size (M)	3
Final smoothing parameter (δ)	3.90625×10^{-4}
Time indicator (η_{time})	$6.936422091432031 \times 10^{-4}$
PA indicator (η_{PA})	0
Smoothing indicator (η_{δ})	$1.2356238434847785 \times 10^{-2}$
Newton iterations (final solve)	6
Newton residual (final solve)	$1.2684496640262498 \times 10^{-10}$
Finalization note	final_resolve

Table: Example 5 (Hypersensitive optimal control): summary of the final adaptive solve.

(i) Optimal Initialization of planes: In the previous example, running the algorithm with $M = 1$ resulted in a singular matrix problem for Newton Method. This was because the initial chosen control produced:

$$H_p = 0 \tag{14}$$

consequently

$$\rho_n = -\frac{1}{2}\langle H_p, H_x\rangle = 0 \implies \bar{r}_n = 0 \tag{15}$$

This later caused no refinement effect and the outer loop passed to the PA and δ refinement routine. To solve this issue we initialize the Hamiltonian with three controls:

- u_{min}
- u_{max}
- u_{mid}

For the original indicator this was not a problem beacuase:

$$\eta_{\text{time},i} = \Delta t_i \Big(\|\nabla_p H_\delta - \nabla_p H_\delta\|_2 + \|\nabla_x H_\delta - \nabla_x H_\delta\|_2 \Big). \tag{16}$$

References

Karlsson, J., Larsson, S., Sandberg, M., Szepessy, A., Tempone, Raul
An error estimate for symplectic euler approximation of optimal control problems.
arxiv, (2014).